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ISAC-Enabled Near-Field High-Mobility UAV Localization: Analysis, Bounds and Algorithm

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***Abstract*—Reliable localization of high-mobility unmanned aerial vehicles (UAVs) is essential for various services in the low-altitude economy. Future mobile networks offer an economical solution for wide-area UAV localization via integrated sensing and communication (ISAC) capabilities. Operating in the near-field region due to larger antenna arrays and higher frequencies, these networks are subject to spherical wave propagation, which induces intrinsic coupling among range, angle, and Doppler shifts. Moreover, this coupling is compounded by the high mobility of UAVs, a critical aspect that existing signal models for near-field localization may not adequately address. In this paper, we analyze the performance degradation in target localization caused by such mobility-induced model misspecification, and propose a localization algorithm for near-field high-mobility UAV target. Specifically, the fundamental limits on achievable parameter estimation accuracy in the presence of model inaccuracies are quantified using the misspecified Cramér-Rao bound (MCRB). Building upon this analysis, we develop a two-stage near-field localization approach to effectively mitigate the parameter coupling exacerbated by high mobility. Initial parameter estimates are efficiently obtained through tensor decomposition, and subsequently refined via quasi-Newton optimization. Simulation results validate the superior localization and velocity estimation accuracy of the proposed algorithm, aligning well with the theoretical performance bounds.**

***Index Terms*—Integrated sensing and communication, misspecified Cramér bound, near-field localization.**

I. Introduction

Unmanned aerial vehicles (UAVs) support critical tasks in low-altitude economy, such as package delivery and environmental monitoring [1]–[3]. Effective execution of these tasks necessitates accurate localization of non-cooperative UAVs with different technologies, including radar, audio, and video [4]–[6]. Existing mobile network, as it evolves towards integrated sensing and communication (ISAC) [7]–[9], offers a cost-effective solution for broad-coverage UAV localization. The localization results, in turn, benefit the ISAC performance through proactive link adaptation in communications and optimized resource allocation in sensing [10], [11]. With the adoption of large antenna arrays and high carrier frequencies within the wireless infrastructure, its operational range of sensing and communication functions extends into the near-field region [12]–[15]. However, localizing UAVs in the near-field region is particularly challenging due to the coupling of range and angle resulting from spherical-wave propagation [16], and the significant dynamic effects associated with inherent high mobility of UAVs.

To estimate the coupled range and angle parameters in near-field region, various methods have been proposed. These methods can be broadly categorized into multi-dimensional joint estimation and multi-stage decoupled estimation. In particular, joint estimation methods obtain range and angle parameters simultaneously, employing various strategies such as dictionary-based approaches [17], [18], subspace-based methods [19], [20], sparse recovery theory [21], [22]. While capable of achieving high accuracy, these methods typically involve extensive multi-dimensional searches or complex iterative optimization processes. In contrast, multi-stage decoupled approaches adopt a step-wise manner, which separates the estimation of angle and range by exploiting the characteristics of receive arrays. Specifically, the covariance matrices [23], [24] or high-order statistics [25]–[27] can be computed to suppress range-dependent terms while preserving angle-dependent phases, enabling angle estimation to be performed separately before range estimation. However, the above methods are primarily designed for localizing stationary targets, and thus do not explicitly account for the high mobility of UAVs.

This high mobility poses challenges to accurate near-field localization, primarily due to two aspects. Firstly, each array element experiences a different Doppler shift due to the curved wavefronts in the near-field [28], [29]. This contrasts with the far-field region, where a uniform Doppler shift is observed by all array elements. Such element-dependent Doppler shift complicates the angle estimation due to the coupling between angle and velocity parameters in the spatial domain. Secondly, the Doppler shifts in the near-field are caused by both radial and transverse velocities, rather than just radial component as in the far-field [30], [31]. The dependence on two velocity components makes it difficult to separate their individual effects, thereby hindering reliable estimation of both velocities and location parameters.

In this regard, simplified models have been employed in near-field localization for high-mobility targets. These simplifications typically involve either neglecting the element-dependent Doppler variation or considering only the radial

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velocity component of target motion. For instance, in the signal model corresponding to a tensor-based near-field localization (TeNFiLoc) method [12], the element-dependent variations in Doppler shifts caused by wavefront curvature is overlooked. More refined models account for the differences in Doppler shifts associated with radial velocity [32], but still disregard the influence of transverse velocity. While transverse velocity has been incorporated into near-field channel modeling in [31], it tailors to predictive beamforming for cooperative users with known geometry, which differs fundamentally from the task of localizing non-cooperative UAVs without prior information. The dependence on incomplete signal models limits the accuracy of existing localization methods in high-mobility near-field environments.

While these approximations lead to degraded localization accuracy, the performance bounds characterizing this degradation have not been explored. Specifically, existing theoretical bounds quantifying the near-field localization performance focus on static scenarios without capturing the target mobility [33]–[37]. More recently, the performance bound for velocity estimation in near-field region is derived and analyzed in [30]. Nevertheless, this analysis focuses solely on the velocity parameters and does not quantify the estimation performance for other crucial localization parameters like range and angle. Consequently, a theoretical analysis to characterize this performance degradation caused by model misspecification in high-mobility scenarios is still lacking.

In this paper, we provide a rigorous analysis of the impact of transverse velocity omission on near-field parameter estimation, quantifying how this model misspecification affects performance limits. Then, we develop a near-field localization algorithm using a signal model that accurately captures the intricate dynamics of high-mobility target. Specifically, to obtain an effective initial estimation of location and velocity parameters, the receives signals are first reformulated into an approximated low-rank tensor. Leveraging canonical polyadic (CP) decomposition, these parameters are inherently retrieved through tensor factorization. Then, the interdependencies embedded in the derivatives with respect to different parameters are leveraged to refine the initial estimates via Broyden–Fletcher–Goldfarb–Shanno (BFGS) quasi-Newton optimization. Extensive simulations are presented to validate both the theoretical analysis and the proposed algorithm.

The main contributions are summarized as follows.

- We formulate a signal model that rigorously incorporates the transverse velocity, explicitly depicting the coupling relationships across temporal, spatial, and frequency domains of the received signals reflected by near-field high-mobility targets.
- To quantify the model misspecification induced by the omission of transverse velocity, we establish the misspecified Cramér–Rao bound (MCRB) for the misspecified model, and compare it with the Cramér–Rao bound (CRB) for the accurate model, validating the performance degradation caused by model approximation.
- Based on the deployed signal model for near-field high-mobility scenarios, we propose a two-stage algorithm, which mitigates the effects of model misspecification and obtains full parameter identifiability. The simulation results indicate that the proposed algorithm achieves enhanced accuracy in near-field target localization and enables the estimation of transverse velocity.

The remainder of the paper is organized as follows. Section II introduces the deployed near-field system model. Section III presents the impact of model misspecification caused by neglecting the transverse velocity. Section IV details the proposed algorithm for near-field high-mobility target localization. We demonstrate the simulation results in Section V, and draw the conclusions in Section VI.

The notation in this paper are summarized in TABLE I.

TABLE I: List of Notation

Symbol	Description
$a, \boldsymbol{a}, \boldsymbol{A}, \boldsymbol{\mathcal{A}}$	Scalar, vector, matrix, and tensor
$[\boldsymbol{a}]_{(i)}$	The i-th element of vector $\boldsymbol{a}$
$[\boldsymbol{A}]_{(i_1,i_2)}$	The (i_1, i_2)-th element of matrix $\boldsymbol{A}$
$[\boldsymbol{A}]_{(i_1:i_2,j_1:j_2)}$	The submatrix corresponding to the i_1-th to i_2-th rows and the j_1-th to j_2-th columns of matrix $\boldsymbol{A}$
$[\boldsymbol{\mathcal{A}}]_{(i_1,i_2,\cdots,i_J)}$	The $(i_1, i_2,\cdots, i_J)$-th element of tensor $\boldsymbol{\mathcal{A}}$
$\mathbb{C}, \mathbb{Z}$	Set of complex numbers, set of intergers
$\jmath$	Imaginary unit
∇	Vector differential operator
$[\![\cdot]\!]$	Canonical polyadic decomposition of a tensor
$\langle\cdot\rangle$	Inner product operator
$\lvert\cdot\rvert$	Absolute value or matrix determinant
$\circ$, $\odot$	Outer product and Hadamard product
$\Re(\cdot), \Im(\cdot)$	Real and imaginary parts of a complex number
$(\cdot)^{\mathrm{T}}, (\cdot)^{\mathrm{H}}$	Transpose and Hermitian transpose
$(\cdot)^{-1}, (\cdot)^{\dagger}$	Inverse and pseudo-inverse
$\lVert\cdot\rVert_{\mathrm{F}}, \lVert\cdot\rVert_2$	Frobenius norm and Euclidean norm
$\mathbb{E}[\cdot]$	Expectation operator
$\mathrm{diag}(\cdot)$, $\mathrm{vec}(\cdot)$	Diagonal and vectorized operators
$\mathbf{I}, \mathbf{1}$	Identity matrix, all-ones vector

II. System Model

In this section, we formulate the signal model for the near-field localization scenario consisting of a base station (BS) and a high-mobility non-cooperative UAV, as shown in Fig. 1.

A. Geometry Description

We consider a BS equipped with an N-element transmit uniform linear array (ULA) that operates in full-duplex mode for both transmission and reception. The inter-element spacing for the ULA is set to $\frac{\lambda}{2}$, where $\lambda = \frac{c}{f_c}$ denotes the signal wavelength, c denotes the light speed, and f_c is the carrier frequency. The ULA is aligned along the x-axis, with its center at the coordinate origin. The coordinate of the n-th array element is denoted by $\boldsymbol{p}_n = [(n-1-\frac{N-1}{2})\frac{\lambda}{2}, 0]^{\mathrm{T}}$.

Consider a UAV located at $\boldsymbol{p} = [\bar{d}\cos\bar{\theta}, \bar{d}\sin\bar{\theta}]^{\mathrm{T}}$ within the Fresnel near-field region of the BS, where $\bar{d}$ denotes the range between the target and the array center, and $\bar{\theta}$ denotes the

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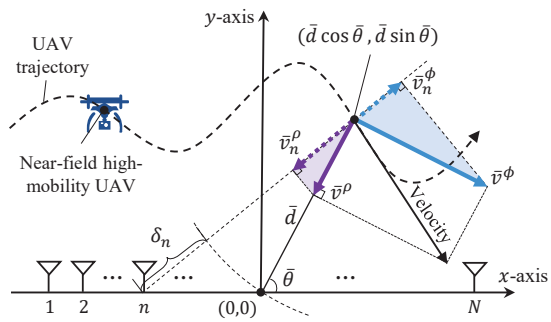

Fig. 1: Illustration of the near-field localization scenario.

direction-of-arrival (DOA) relative to the x-axis. For notational consistency, we denote the true value for the corresponding parameter of interest with an overline, i.e. $\bar{(\cdot)}$, throughout this paper. The range between the target and the n-th array element can be represented as

$$r_n = \|\boldsymbol{p}_n - \boldsymbol{p}\| = \sqrt{\bar{d}^2 + \frac{\dot{n}^2}{4}\lambda^2 - \bar{d}\dot{n}\lambda\cos\bar{\theta}}, \tag{1}$$

where $\dot{n} = n - \frac{N+1}{2}$ represents the shifted element index with respect to the array center.

The motion of the UAV is depicted by the radial velocity $\bar{v}^{\rho}$ in the direction of propagation and the transverse velocity $\bar{v}^{\phi}$ orthogonal to the radial direction. Under near-field conditions, both radial and transverse velocities contribute to the Doppler frequency shifts due to their spatially varying projections on the spherical wavefronts. Specifically, the equivalent velocity observed at the n-th array element is given by

$$\nu_n = \bar{\nu}_n^{\rho} + \bar{\nu}_n^{\phi}, \tag{2}$$

where

$$\bar{\nu}_n^{\rho} = \frac{\bar{d} - \dot{n}\frac{\lambda}{2}\cos\bar{\theta}}{r_n}\bar{v}^{\rho}, \quad \bar{\nu}_n^{\phi} = \frac{\dot{n}\frac{\lambda}{2}\sin\bar{\theta}}{r_n}\bar{v}^{\phi}. \tag{3}$$

represent the contributions from the radial velocity $\bar{v}^{\rho}$ and the transverse velocity $\bar{v}^{\phi}$, respectively.

B. Received Signal Model

To actively sense the non-cooperative UAV, the BS transmits orthogonal frequency division multiplexing (OFDM) waveforms comprising K narrowband subcarriers over an observation period with M consecutive symbol durations. We assume the UAV's location and velocity parameters are quasi-static within this observation interval. This is a common assumption in channel modeling, justified by the fact that these parameters remain effectively constant over such short durations [38].

After target reflection, the baseband equivalent received signals $\mathrm{y}_{k,m,n}$ corresponding to the k-th ($k = 1, 2, \cdots, K$) subcarrier and the m-th ($m = 1, 2, \cdots, M$) symbol at the n-th ($n = 1, 2, \cdots, N$) receive array element can be represented as

$$\mathrm{y}_{k,m,n} = \sum_{\tilde{n}=1}^{N} \dot{\beta} e^{-\jmath 2\pi \frac{f_c + k\Delta}{c}(r_n + r_{\tilde{n}})} e^{\jmath 2\pi m T_{\mathrm{s}}(\nu_n + \nu_{\tilde{n}})} s_{k,m,\tilde{n}} + n_{k,m,n}, \tag{4}$$

where $\tilde{n}$ is the index of transmit array element, $\dot{\beta}$ is the reflection coefficient, Δ denotes the subcarrier spacing, T_{s} denotes the symbol duration, $s_{k,m,n}$ is the complex transmitted symbol, and $n_{k,m,n} \sim \mathcal{CN}(0, \sigma^2)$ is the complex Gaussian noise with zero mean and variance σ^2. Following the frequency-flat transmission scheme in [39], [40], we set the identical transmitted symbol across all subcarriers such that $s_{k,m,\tilde{n}} = s_{m,\tilde{n}}, \forall k = 1, 2, \cdots, K$. Note that, the received signals exhibits inherent coupling characteristics due to the near-field and high-mobility nature of the scenario. Particularly, the first phase term $e^{-\jmath 2\pi \frac{f_c + k\Delta}{c}(r_n + r_{\tilde{n}})}$ in (4) leads to spatial-frequency coupling, while the second phase term $e^{\jmath 2\pi m T_{\mathrm{s}}(\nu_n + \nu_{\tilde{n}})}$ in (4) induces spatial-temporal coupling.

Under the narrowband signal assumption $\Delta \ll f_c$, we have $\frac{f_c + k\Delta}{c} \approx \frac{1}{\lambda}$. Accordingly, the range-related phase term in (4) can be approximated as

$$e^{-\jmath 2\pi \frac{f_c + k\Delta}{c}(r_n + r_{\tilde{n}})} \approx e^{-\jmath \frac{2\pi}{\lambda}(\delta_n + \delta_{\tilde{n}})} \cdot e^{-\jmath 2\pi \frac{f_c + k\Delta}{c} 2\bar{d}}, \tag{5}$$

where the total distance r_n is decomposed into the range $\bar{d}$ away from the array center and an element-wise offset

$$\delta_n \triangleq r_n - \bar{d} = \sqrt{\bar{d}^2 + \frac{\dot{n}^2}{4}\lambda^2 - \bar{d}\dot{n}\lambda\cos\bar{\theta}} - \bar{d}. \tag{6}$$

To express $\mathrm{y}_{k,m,n}$ in (4) into a more compact vectorized form, we define the near-field spatial and temporal steering vectors as

$$\boldsymbol{a}(\bar{\theta}, \bar{d}) = \left[e^{-\jmath \frac{2\pi}{\lambda}\delta_1}, e^{-\jmath \frac{2\pi}{\lambda}\delta_2}, \cdots, e^{-\jmath \frac{2\pi}{\lambda}\delta_N}\right]^{\mathrm{T}} \tag{7}$$

and

$$\boldsymbol{d}_m(\bar{v}^{\rho}, \bar{v}^{\phi}, \bar{\theta}, \bar{d}) = \left[e^{\jmath \frac{2\pi}{\lambda}\nu_1 m T_{\mathrm{s}}}, e^{\jmath \frac{2\pi}{\lambda}\nu_2 m T_{\mathrm{s}}}, \cdots, e^{\jmath \frac{2\pi}{\lambda}\nu_N m T_{\mathrm{s}}}\right]^{\mathrm{T}}, \tag{8}$$

respectively. By aggregating the received signals $\mathrm{y}_{k,m,n}$ across all array elements as $\mathbf{y}_{k,m} = [\mathrm{y}_{k,m,1}, \mathrm{y}_{k,m,2}, \cdots, \mathrm{y}_{k,m,N}]^{\mathrm{T}}$, and substituting (7) and (8) into (4), the received signal vector $\mathbf{y}_{k,m}$ can be compactly expressed as

$$\mathbf{y}_{k,m} = \bar{\beta} e^{-\jmath 2\pi k\Delta \frac{2\bar{d}}{c}} \left(\boldsymbol{A}(\bar{\theta}, \bar{d}) \odot \boldsymbol{D}_m(\bar{v}^{\rho}, \bar{v}^{\phi}, \bar{\theta}, \bar{d})\right) \mathbf{s}_m + \mathbf{n}_{k,m}. \tag{9}$$

Here, $\bar{\beta} = \dot{\beta} e^{-\jmath 2\pi \frac{f_c}{c} 2\bar{d}}$ captures both the roundtrip attenuation and the reflection coefficient of the target,

$$\boldsymbol{A}(\bar{\theta}, \bar{d}) = \boldsymbol{a}\,\boldsymbol{a}^{\mathrm{T}}, \quad \boldsymbol{D}_m(\bar{v}^{\rho}, \bar{v}^{\phi}, \bar{\theta}, \bar{d}) = \boldsymbol{d}_m\,\boldsymbol{d}_m^{\mathrm{T}} \tag{10}$$

are respectively the spatial and temporal steering matrices, $\mathbf{s}_m = [s_{m,1}, s_{m,2}, \cdots, s_{m,N}]^{\mathrm{T}}$ denotes the transmitted symbol vector, and $\mathbf{n}_{k,m} \sim \mathcal{CN}(0, \sigma^2\mathbf{I})$ represents the additive complex Gaussian noise.

C. Parameters Coupling

For real-time localization of UAVs, it is necessary to estimate the parameters $\{d, \theta, v^{\rho}, v^{\phi}\}$ from the received signal vector $\mathbf{y}_{k,m}$ in (9). However, this task is challenging in the near-field due to parameters coupling across the spatial, frequency, and temporal domains. We identify two primary types of coupling in the following remark.

Remark 1: First, the spatial steering matrix $\boldsymbol{A}(\theta, d)$ in (10) nonlinearly couples angle θ and range d through near-field phase terms δ_n in (6), leading to spatial-frequency interdependence. Moreover, the temporal steering matrix $\boldsymbol{D}_m(v^{\rho}, v^{\phi})$ in (10) induces spatial-temporal coupling via ν_n in (2), since the

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projections of v^ρ and v^ϕ spatially vary across different array elements.

Existing methods for near-field non-cooperative target localization typically either consider static scenarios (i.e., $v^\rho = 0, v^\phi = 0$) or assume the target moves along the radial direction (i.e., $v^\rho \neq 0$, $v^\phi = 0$). These constraints inevitably lead to model mismatch in the more general case of $v^\phi \neq 0$. In this regard, theoretical bounds are derived to quantify the performance degradation induced by $v^\phi = 0$ in Section III. Furthermore, a two-stage algorithm is proposed in Section IV to resolve the triple-domain coupling under near-field high-mobility conditions.

III. Model Misspecification associated with Transverse Mobility

This section leverages misspecified estimation theory to evaluate the performance degradation caused by the ignorance of transverse mobility. Specifically, we establish the CRB for the true model and derive the MCRB for the misspecified model, respectively.

Before deriving theoretical performance bounds, we present the probability density functions (PDFs) of the true and misspecified signal models. Specifically, by denoting $\mathbf{y}_k = [\mathbf{y}_{k,1}, \mathbf{y}_{k,2}, \cdots, \mathbf{y}_{k,M}]^{\mathrm{T}}$, $\boldsymbol{y} = [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_K]^{\mathrm{T}}$, we have

$$\begin{aligned} p(\boldsymbol{y}\,|\,\boldsymbol{\eta}) &= \prod_{k,m} p(\mathbf{y}_{k,m}\,|\,\boldsymbol{\eta}) \\ &= \prod_{k,m} \frac{1}{(\pi\sigma^2)^N} \exp\Big(-\frac{1}{\sigma^2}\big\|\mathbf{y}_{k,m} - \boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})\big\|^2\Big), \end{aligned} \tag{11}$$

where

$$\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta}) = \beta e^{-\jmath 2\pi k \Delta \frac{2d}{c}} \big(\boldsymbol{A}(\theta, d) \odot \boldsymbol{D}_m(v^\rho, v^\phi, \theta, d)\big)\, \mathbf{s}_m \tag{12}$$

represents the noise-free counterpart of $\mathbf{y}_{k,m}$ in (9), $\boldsymbol{\eta} = [d, \theta, v^\rho, v^\phi, \beta_{\mathrm{R}}, \beta_{\mathrm{I}}]^{\mathrm{T}}$ denotes the parameter vector, $\beta_{\mathrm{R}} = \Re(\beta)$ and $\beta_{\mathrm{I}} = \Im(\beta)$ denote the real and imaginary parts of β, respectively. Then, the PDFs of the true signal model and the misspecified model are respectively presented as follows:

- True model: The PDF is denoted by $p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})$, where $\bar{\boldsymbol{\eta}} = [\bar{d}, \bar{\theta}, \bar{v}^\rho, \bar{v}^\phi, \Re(\bar{\beta}), \Im(\bar{\beta})]^{\mathrm{T}}$ represents the true parameter vector with the true transverse velocity $\bar{v}^\phi$.
- Misspecified model: The PDF is denoted by $\tilde{p}(\boldsymbol{y} \mid \tilde{\boldsymbol{\eta}})$, where $\tilde{\boldsymbol{\eta}} = [d, \theta, v^\rho, v^\phi \equiv 0, \beta_{\mathrm{R}}, \beta_{\mathrm{I}}]^{\mathrm{T}}$ is the misspecified parameter vector, which assumes the transverse velocity v^ϕ to be 0.

A. Derivation of CRB for the True Model

Based on the true parameter vector $\bar{\boldsymbol{\eta}}$ and the corresponding PDF $p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})$, the Fisher information matrix (FIM) of all the parameters can be formulated as

$$\boldsymbol{F}(\bar{\boldsymbol{\eta}}) = -\mathbb{E}_p\Big[\nabla^2_{\bar{\boldsymbol{\eta}}}\, \ln p(\boldsymbol{y} \mid \bar{\boldsymbol{\eta}})\Big], \tag{13}$$

where $\mathbb{E}_p$ denotes the expectation operator under the PDF $p(\boldsymbol{y} \mid \bar{\boldsymbol{\eta}})$ of true signal model. By substituting (11) into (13), the (i,j)-th element in $\boldsymbol{F}(\bar{\boldsymbol{\eta}})$ is given by

$$\begin{aligned} [\boldsymbol{F}(\bar{\boldsymbol{\eta}})]_{(i,j)} = \frac{-1}{\sigma^2} \sum_{k,m} \mathbb{E}_p \Bigg[2\Re\Bigg\{ \Big(\frac{\partial^2 \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial\eta_j \partial\eta_i}\Big)^{\mathrm{H}} (\mathbf{y}_{k,m} - \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})) \Bigg\} \\ -2\Re\Bigg\{ \Big(\frac{\partial \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial\eta_i}\Big)^{\mathrm{H}} \Big(\frac{\partial \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial\eta_j}\Big) \Bigg\} \Bigg]. \end{aligned} \tag{14}$$

Noting that $\mathbb{E}_p[\mathbf{y}_{k,m} - \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})] = 0$, we have

$$\boldsymbol{F}(\bar{\boldsymbol{\eta}}) = \frac{2}{\sigma^2} \sum_{k=1}^{K} \sum_{m=1}^{M} \Re\Bigg\{ \Big(\frac{\partial \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial \bar{\boldsymbol{\eta}}}\Big)^{\mathrm{H}} \frac{\partial \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial \bar{\boldsymbol{\eta}}} \Bigg\}, \tag{15}$$

where $\frac{\partial \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\partial \bar{\boldsymbol{\eta}}}$ is the Jacobian matrix of $\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})$ with respect to $\boldsymbol{\eta}$, evaluated at $\bar{\boldsymbol{\eta}}$. Then, the CRB matrix of the true model can be obtained as

$$\mathrm{CRB}(\bar{\boldsymbol{\eta}}) = \boldsymbol{F}^{-1}(\bar{\boldsymbol{\eta}}). \tag{16}$$

The detailed derivatives of $\frac{\partial \boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}}$ are presented in Appendix A.

In the task of estimating the location and velocities of the UAV target, the parameter vector can be partitioned into parameters of interest $\{d, \theta, v^\rho, v^\phi\}$ and uninterested nuisance parameters $\{\beta_{\mathrm{R}}, \beta_{\mathrm{I}}\}$. Accordingly, the FIM $\boldsymbol{F}(\bar{\boldsymbol{\eta}})$ can be divided into a 2×2 block matrix

$$\boldsymbol{F}(\bar{\boldsymbol{\eta}}) = \begin{bmatrix} \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} & \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} \\ \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} & \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} \end{bmatrix}, \tag{17}$$

where $\bar{\boldsymbol{\eta}}_{\mathrm{i}} = [\bar{d}, \bar{\theta}, \bar{v}^\rho, \bar{v}^\phi]^{\mathrm{T}}$, and $\bar{\boldsymbol{\eta}}_{\mathrm{u}} = [\bar{\beta}_{\mathrm{R}}, \bar{\beta}_{\mathrm{I}}]^{\mathrm{T}}$. To derive the CRBs for the parameters of interest, the Schur complement corresponding to the partition in (17) is [41]

$$\boldsymbol{F}'_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} = \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} - \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} \boldsymbol{F}^{-1}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} \boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}}. \tag{18}$$

Let $\boldsymbol{\mu}_{\mathrm{i},k,m}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) = \frac{\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})}{\bar{\beta}}$, $\boldsymbol{\Xi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) = [\boldsymbol{\mu}_{\mathrm{i},1,1}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}), \cdots, \boldsymbol{\mu}_{\mathrm{i},K,1}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}), \boldsymbol{\mu}_{\mathrm{i},1,2}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}), \cdots, \boldsymbol{\mu}_{\mathrm{i},K,M}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})]^{\mathrm{T}}$, $\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) = \mathrm{vec}(\boldsymbol{\Xi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}))$, and $\boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) = [\frac{\partial \boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})}{\partial d}, \frac{\partial \boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})}{\partial \theta}, \frac{\partial \boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})}{\partial v^\rho}, \frac{\partial \boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})}{\partial v^\phi}]$. The sub-blocks of FIM can then be expressed as

$$\boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} = \frac{2|\bar{\beta}|^2}{\sigma^2} \Re\left\{ \boldsymbol{\Pi}^{\mathrm{H}}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \right\}, \tag{19}$$

$$\boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} = \frac{2||\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})||^2}{\sigma^2} \mathbf{I}, \tag{20}$$

$$\boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} = \frac{2}{\sigma^2} \begin{bmatrix} \Re\{\bar{\beta} \boldsymbol{\varrho}^{\mathrm{H}}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})\} \\ \Im\{\bar{\beta} \boldsymbol{\varrho}^{\mathrm{H}}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})\} \end{bmatrix}, \tag{21}$$

and $\boldsymbol{F}_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{u}}} = \boldsymbol{F}^{\mathrm{T}}_{\bar{\boldsymbol{\eta}}_{\mathrm{u}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}}$. Substituting these expressions into (18), and after some algebraic manipulation, the equivalent FIM for $\boldsymbol{\eta}_{\mathrm{i}}$ can be written in a compact form involving an orthogonal projection matrix:

$$\boldsymbol{F}'_{\bar{\boldsymbol{\eta}}_{\mathrm{i}}\bar{\boldsymbol{\eta}}_{\mathrm{i}}} = \frac{2|\bar{\beta}|^2}{\sigma^2} \Re\left\{ (\boldsymbol{\Pi}^{\mathrm{H}}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})) \boldsymbol{P}^{\perp}_{\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})} \boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \right\}, \tag{22}$$

where

$$\boldsymbol{P}^{\perp}_{\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})} = \mathbf{I} - \frac{\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}}) \boldsymbol{\varrho}^{\mathrm{H}}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})}{||\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})||^2} \tag{23}$$

denotes the projection matrix onto the subspace orthogonal to the stacked signal structure vector $\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\mathrm{i}})$. Finally, the CRB for

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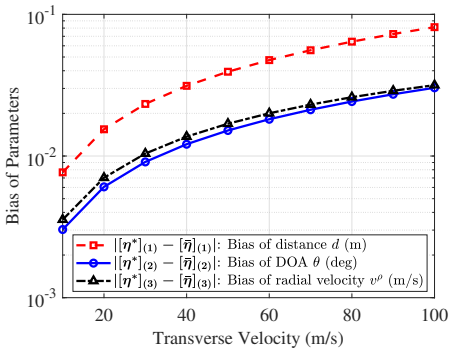

Fig. 2: Bias of the pseudo-true range, DOA, and radial velocity from the corresponding true parameters, where the system setting and the true parameters are given in Section V.

the parameters of interest $\boldsymbol{\eta}_{\rm i}$ can be represented as

$$\mathrm{CRB}(\bar{\boldsymbol{\eta}}_{\rm i}) = \left(\boldsymbol{F}'_{\bar{\boldsymbol{\eta}}_{\rm i}\bar{\boldsymbol{\eta}}_{\rm i}}\right)^{-1} = \left(\frac{2|\bar{\beta}|^2}{\sigma^2}\Re\left\{\boldsymbol{\Pi}^{\rm H}(\bar{\boldsymbol{\eta}}_{\rm i})\boldsymbol{P}^{\perp}_{\boldsymbol{\varrho}(\bar{\boldsymbol{\eta}}_{\rm i})}\boldsymbol{\Pi}(\bar{\boldsymbol{\eta}}_{\rm i})\right\}\right)^{-1}. \tag{24}$$

The CRB derived under the true model quantifies the best achievable estimation accuracy of location and velocity parameters. However, this bound becomes loose when the model is misspecified, as it fails to account for the resulting systematic bias. This motivates the use of alternative bounds that remain valid under model misspecification.

B. Derivation of MCRB for the Misspecified Model

In the following, we derive the MCRB to characterize estimation performance in the presence of model misspecification, capturing both the stochastic uncertainty and systematic bias.

1) Determination of the Pseudo-True Parameters: According to the definition of MCRB [42], [43], we first determine the pseudo-true parameter vector

$$\boldsymbol{\eta}^* = \arg\min_{\tilde{\boldsymbol{\eta}}} D_{\rm KL}\Big(p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})\,\big\|\,\tilde{p}(\boldsymbol{y}\,|\,\tilde{\boldsymbol{\eta}})\Big), \tag{25}$$

which minimizes the Kullback-Leibler (KL) divergence

$$D_{\rm KL}\Big(p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})\,\big\|\,\tilde{p}(\boldsymbol{y}\,|\,\tilde{\boldsymbol{\eta}})\Big) \triangleq \int p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})\ln\frac{p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})}{\tilde{p}(\boldsymbol{y}\,|\,\tilde{\boldsymbol{\eta}})}d\boldsymbol{y} \tag{26}$$

between the true and misspecified PDFs in (11).

Proposition 1: The pseudo-true parameter vector $\boldsymbol{\eta}^*$ can be obtained by minimizing the difference between the noise-free signals corresponding to $\bar{\boldsymbol{\eta}}$ and $\tilde{\boldsymbol{\eta}}$, i.e.,

$$\boldsymbol{\eta}^* = \arg\min_{\tilde{\boldsymbol{\eta}}} \sum_{k,m} \|\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}}) - \boldsymbol{\mu}_{k,m}(\tilde{\boldsymbol{\eta}})\|^2. \tag{27}$$

Proof: By substituting (11) into (26), $D_{\rm KL}\left(p\,\|\,\tilde{p}\right)$ can be rewritten as

$$D_{\rm KL}\left(p\,\|\,\tilde{p}\right) = \sum_{k,m}\int p(\mathbf{y}_{k,m}\,|\,\bar{\boldsymbol{\eta}})\ln\frac{p(\mathbf{y}_{k,m}\,|\,\bar{\boldsymbol{\eta}})}{\tilde{p}(\mathbf{y}_{k,m}\,|\,\tilde{\boldsymbol{\eta}})}d\mathbf{y}_{k,m}, \tag{28}$$

where $\mathbf{y}_{k,m}\mid\bar{\boldsymbol{\eta}} \sim \mathcal{CN}(\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}}),\sigma^2\mathbf{I})$ and $\mathbf{y}_{k,m}\mid\tilde{\boldsymbol{\eta}} \sim \mathcal{CN}(\boldsymbol{\mu}_{k,m}(\tilde{\boldsymbol{\eta}}),\sigma^2\mathbf{I})$ follow complex Gaussian distribution, respectively. According to the standard results on the KL divergence of Gaussian distributions [44], a closed-form expression of $D_{\rm KL}\left(p\,\|\,\tilde{p}\right)$ is given by

$$D_{\rm KL}\left(p\,\|\,\tilde{p}\right) = \sum_{k,m}\frac{1}{\sigma^2}\|\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}}) - \boldsymbol{\mu}_{k,m}(\tilde{\boldsymbol{\eta}})\|^2. \tag{29}$$

Substituting (29) into (25), the pseudo-true parameter vector $\boldsymbol{\eta}^*$ can be calculated as in (27). ■

To illustrate the impact of the transverse velocity v^ϕ on the determination of pseudo-true parameters, the bias between true and pseudo parameters is depicted in Fig. 2. As transverse velocity v^ϕ increases, the deviation between $\boldsymbol{\eta}^*$ and $\bar{\boldsymbol{\eta}}$ grows accordingly, which deteriorates the achievable estimation performance, as will be further discussed in the subsequent derivation of the MCRB.

2) Derivation of MCRB: Based on the pseudo-true parameter vector $\boldsymbol{\eta}^*$, we define the negative expected Hessian and the gradient covariance matrix of the misspecified model as

$$\boldsymbol{C}(\boldsymbol{\eta}^*) = -\,\mathbb{E}_p\Big[\nabla^2_{\boldsymbol{\eta}}\,\ln\tilde{p}(\boldsymbol{y}\mid\boldsymbol{\eta})\big|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}\Big] \tag{30}$$

and

$$\boldsymbol{B}(\boldsymbol{\eta}^*) = \mathbb{E}_p\Big[\nabla_{\boldsymbol{\eta}}\,\ln\tilde{p}(\boldsymbol{y}\,|\,\boldsymbol{\eta})\,\nabla^{\rm T}_{\boldsymbol{\eta}}\,\ln\tilde{p}(\boldsymbol{y}\,|\,\boldsymbol{\eta})\big|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}\Big], \tag{31}$$

respectively. Then, the MCRB matrix for an estimator of $\bar{\boldsymbol{\eta}}$ is given by

$$\mathrm{MCRB}(\bar{\boldsymbol{\eta}}) = \boldsymbol{C}^{-1}(\boldsymbol{\eta}^*)\boldsymbol{B}(\boldsymbol{\eta}^*)\boldsymbol{C}^{-1}(\boldsymbol{\eta}^*) + (\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})^{\rm T}, \tag{32}$$

where $(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})^{\rm T}$ is the squared asymptotic bias term. The detailed calculation of $\boldsymbol{C}(\boldsymbol{\eta}^*)$ and $\boldsymbol{B}(\boldsymbol{\eta}^*)$ are presented in Appendix B. When the model is correctly specified (i.e., $p=\tilde{p}$), the pseudo-true parameter $\boldsymbol{\eta}^*$ coincides with the true parameter $\bar{\boldsymbol{\eta}}$. Consequently, the bias term $(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})^{\rm T}$ becomes $\mathbf{0}$. Meanwhile, the matrices $\boldsymbol{B}(\boldsymbol{\eta}^*)$ and $\boldsymbol{C}(\boldsymbol{\eta}^*)$ become equal to the FIM, i.e., $\boldsymbol{B}(\boldsymbol{\eta}^*)=\boldsymbol{C}(\boldsymbol{\eta}^*)=\boldsymbol{F}(\bar{\boldsymbol{\eta}})$. Thus, the MCRB reduces to $\boldsymbol{F}(\bar{\boldsymbol{\eta}})$, which is the classical CRB.

The MCRB in (32) serves as a lower bound on the mean squared error (MSE) matrix for estimators derived under model misspecification [45]:

$$\mathrm{MSE}(\bar{\boldsymbol{\eta}}) \succeq \mathrm{MCRB}(\bar{\boldsymbol{\eta}}). \tag{33}$$

The additive bias term $(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})^{\rm T}$ reveals that a larger deviation between $\boldsymbol{\eta}^*$ and $\bar{\boldsymbol{\eta}}$ leads to increased MSE under model misspecification. This component cannot be mitigated by merely increasing measurement diversity (e.g., number of symbol durations M or number of subcarriers K).

Remark 2: Unlike the CRB in (24) that monotonically decreases with increasing K and M due to enhanced Fisher information, the MCRB do not necessarily follows this decreasing trend. This is because increasing M and K not only provides more independent observations but also amplifies the discrepancy between the true and misspecified models. To be specific, for the MCRB, the growth of the cross-terms involving residual errors $\boldsymbol{\varepsilon}_{k,m}(\boldsymbol{\eta}) = \boldsymbol{\mu}_{k,m}(\boldsymbol{\eta}) - \boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})$ in matrix $\boldsymbol{B}(\boldsymbol{\eta}^*)$ in (64) may potentially counteract the anticipated information gain. Meanwhile, the systematic bias term $(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})(\boldsymbol{\eta}^*-\bar{\boldsymbol{\eta}})^{\rm T}$ in (32) can accumulate with increasing M and K, ultimately dominating the overall error bound.

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The analysis and bounds in this section demonstrate that neglecting transverse velocity components in high-mobility scenarios leads to systematic bias in parameter estimation. This underscores the importance of explicitly modeling transverse motion to accurately capture the dynamic state of the UAV and reduce estimation errors associated with location parameters.

IV. Parameter Estimation under Motion-Induced Spatial-Temporal Coupling

To estimate the location and velocity parameters without neglecting the transverse velocity v^{ϕ} in signal model, we formulate the following maximum likelihood problem

$$\begin{aligned} \min_{\boldsymbol{\eta}=[d,\theta,v^{\rho},v^{\phi},\beta_{\mathrm{R}},\beta_{\mathrm{I}}]^{\mathrm{T}}} \quad & J(\boldsymbol{\eta}) \triangleq \sum_{k=1}^{K}\sum_{m=1}^{M} \left\|\mathbf{y}_{k,m}-\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})\right\|_{\mathrm{F}}^{2}, \\ \text{s.t.} \quad & 0<d\leq \frac{c}{2\Delta f}, \quad |v^{\rho}|\leq \frac{\lambda}{4T_{\mathrm{s}}}. \end{aligned} \tag{34}$$

where the constraints delineate the maximum unambiguous range $\frac{c}{2\Delta}$ and the maximum unambiguous velocity $\frac{\lambda}{4T_{\mathrm{s}}}$. Due to the model identifiability of $\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})$, solving (34) leads to an unbiased estimation of $\bar{\boldsymbol{\eta}}$.

The problem in (34) is inherently non-convex and exhibits multiple local extrema owing to the non-convex substructures in $J(\boldsymbol{\eta})$, including nonlinear complex exponentials, squared magnitudes, and fractional components. As a result, the optimization outcome is sensitive to initialization. In cooperative scenarios, prior knowledge of location parameters such as $\bar{d}$ and $\bar{\theta}$ may be available in initialization [31]. However, in our case with a non-cooperative UAV, such prior information is unavailable, and random initialization can lead to poor convergence or suboptimal solutions.

In this regard, we develop a two-stage algorithm, where coarse estimates are first obtained under relaxed kinematic constraints, and then refined via quasi-Newton optimization to mitigate systematic bias and enhance estimation accuracy.

A. Coarse Initialization via CP Decomposition

To get a reliable initialization, we exploit the intrinsic multi-dimensional structure of the received signal by adopting a tensor-based formulation. Specifically, the received vectors are arranged into a high-order tensor to preserve correlations across various domains, enabling low-complexity parameter extraction via CP decomposition.

1) Formulation of Approximated Tensor Signal Model: In particular, $[\mathbf{y}_{k,m}]_{(n)}$ in (4) across all indices n, m, k are organized into a third-order tensor as

$$[\boldsymbol{\mathcal{Y}}]_{(n,m,k)} = [\mathbf{y}_{k,m}]_{(n)}. \tag{35}$$

Due to the intricate coupling among space, time, and frequency, the received signal tensor $\boldsymbol{\mathcal{Y}}$ lacks an explicit separable structure, making it difficult to extract low-rank components associated with individual parameters. To enable tractable low-rank tensor decomposition, such as CP decomposition, we introduce the following approximations to promote a low-rank representation of $\boldsymbol{\mathcal{Y}}$.

i. The Doppler shift component caused by transverse velocity $\bar{v}^{\phi}$ is neglected in the Doppler shift vector defined in (8), i.e., $\boldsymbol{d}_m(\bar{v}^{\rho},\bar{v}^{\phi}) \approx \boldsymbol{d}_m(\bar{v}^{\rho},0)$, since $\nu_n^{\rho} \gg \nu_n^{\phi}$ typically holds in (3), leading to $\nu_n \approx \nu_n^{\rho}$ in (2).

ii. Variations in the projections of the radial velocity across array elements are disregarded, i.e., $\nu_n^{\rho} \approx \bar{v}^{\rho}$ in (3), leading to $\boldsymbol{d}_m(\bar{v}^{\rho},0) \approx e^{\jmath\frac{2\pi}{\lambda}\bar{v}^{\rho}mT_{\mathrm{s}}}\mathbf{1}$ in (8).

These approximations enable efficient parameter extraction via CP decomposition, while the inaccuracies introduced by this process will be compensated for in the subsequent refinement stage. Accordingly, the received signal vector $\mathbf{y}_{k,m}$ in (9) can be rewritten as

$$\mathbf{y}_{k,m} \approx \bar{\beta} e^{-\jmath 2\pi k\Delta\frac{2\bar{d}}{c}} e^{\jmath\frac{2\pi}{\lambda}2\bar{v}^{\rho}mT_{\mathrm{s}}} \boldsymbol{A}(\bar{\theta},\bar{d})\ \mathbf{s}_m + \mathbf{n}_{k,m}. \tag{36}$$

By substituting (36) into (35), an approximated signal tensor $\tilde{\boldsymbol{\mathcal{Y}}}$ is yielded, which admits a rank-one decomposition as

$$\tilde{\boldsymbol{\mathcal{Y}}} = \bar{\beta}\boldsymbol{a}(\bar{\theta},\bar{d}) \circ \boldsymbol{u}(\bar{d}) \circ \left[\boldsymbol{S}\boldsymbol{a}(\bar{\theta},\bar{d}) \odot \boldsymbol{f}(\bar{v}^{\rho})\right] + \boldsymbol{\mathcal{N}}. \tag{37}$$

Here, $\boldsymbol{u}(\bar{d}) = [e^{-\jmath 2\pi\Delta\frac{2\bar{d}}{c}}, e^{-\jmath 2\pi 2\Delta\frac{2\bar{d}}{c}}, \cdots, e^{-\jmath 2\pi K\Delta\frac{2\bar{d}}{c}}]^{\mathrm{T}}$, $\boldsymbol{f}(\bar{v}^{\rho}) = [e^{\jmath\frac{2\pi}{\lambda}2\bar{v}^{\rho}T_{\mathrm{s}}}, e^{\jmath\frac{2\pi}{\lambda}2\bar{v}^{\rho}2T_{\mathrm{s}}}, \cdots, e^{\jmath\frac{2\pi}{\lambda}2\bar{v}^{\rho}MT_{\mathrm{s}}}]^{\mathrm{T}}$, $\boldsymbol{S} = [\mathbf{s}_1, \mathbf{s}_2, \cdots, \mathbf{s}_M]^{\mathrm{T}}$ and $\boldsymbol{\mathcal{N}}$ denotes the noise tensor. Note that, the goal of this stage is to provide a robust and fast initialization rather than achieving ultimate precision. Therefore, while other methods are applicable [32], [46], the CP decomposition-based approach is selected for its superior trade-off between estimation accuracy and computational complexity.

2) CP Decomposition for Parameter Estimation: Building upon the low-rank tensor signal model $\tilde{\boldsymbol{\mathcal{Y}}}$, the CP decomposition is employed to extract initial parameter estimates $\{\hat{d},\hat{\theta},\hat{v}^{\rho},\hat{\beta}\}$. To this end, the parameter estimation task is reformulated as a tensor factorization problem, where $\tilde{\boldsymbol{\mathcal{Y}}}$ is approximated by the outer product of three latent factor vectors $\boldsymbol{g}_1 = \beta\boldsymbol{a}(\theta,d)$, $\boldsymbol{g}_2 = \boldsymbol{u}(d)$, and $\boldsymbol{g}_3 = \boldsymbol{S}\boldsymbol{a}(\theta,d) \odot \boldsymbol{f}(v^{\rho})$. These factor vectors are jointly estimated through the following least-squares problem

$$\min_{\hat{\boldsymbol{g}}_1,\hat{\boldsymbol{g}}_2,\hat{\boldsymbol{g}}_3} \left\|\tilde{\boldsymbol{\mathcal{Y}}} - [\![\hat{\boldsymbol{g}}_1,\hat{\boldsymbol{g}}_2,\hat{\boldsymbol{g}}_3]\!]\right\|_{\mathrm{F}}^{2}, \tag{38}$$

where $\hat{\boldsymbol{g}}_1$, $\hat{\boldsymbol{g}}_2$, $\hat{\boldsymbol{g}}_3$ are the estimates of $\boldsymbol{g}_1$, $\boldsymbol{g}_2$, and $\boldsymbol{g}_3$, respectively. To enhance estimation performance, prior structural information of the factor vectors can be exploited. As an illustrative example, a constrained alternating least squares (ALS) algorithm is employed in this paper, where the Vandermonde structure is imposed on the factor vector $\hat{\boldsymbol{g}}_2$. In general, the CP decomposition in (38) can be solved by various methods, such as standard ALS algorithm and other structure-constrained tensor decomposition algorithms [47], [48].

To determine the location parameters of the target, the range between the target and all the array element r_n can be estimated by $\hat{r}_n = \angle([\hat{\boldsymbol{g}}_1]_{(n)})$. Meanwhile, the range d between the target and the array center is estimated by

$$\hat{d} = \arg\max_{d} \frac{\left|\hat{\boldsymbol{g}}_2^{\mathrm{H}}\boldsymbol{u}(d)\right|^2}{\left\|\hat{\boldsymbol{g}}_2^{\mathrm{H}}\right\|_2^2 \left\|\boldsymbol{u}(d)\right\|_2^2}. \tag{39}$$

According to the near-field geometry relationship between d and θ demonstrated in (1), the estimate of DOA θ can be

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obtained as

$$\hat{\theta} = \arccos\left(\sum_{n=1}^{N} \frac{\hat{d}^2 + (n-1-\frac{N-1}{2})^2(\frac{\lambda}{2})^2 - \hat{r}_n^2}{(n-1-\frac{N-1}{2})\lambda\hat{d}}\right). \quad (40)$$

Based on the estimates $\hat{d}$ and $\hat{\theta}$, the radial velocity can be estimated by

$$\hat{v}^{\rho} = \arg\max_{v^{\rho}} \frac{\left|\hat{\boldsymbol{g}}_3^{\mathrm{H}}\left[\boldsymbol{f}(v^{\rho})\odot\left(\boldsymbol{S}\boldsymbol{a}(\hat{\theta},\hat{d})\right)\right]\right|^2}{\left\|\hat{\boldsymbol{g}}_3^{\mathrm{H}}\right\|_2^2\left\|\boldsymbol{f}(v^{\rho})\odot\left(\boldsymbol{S}\boldsymbol{a}(\hat{\theta},\hat{d})\right)\right\|_2^2}. \quad (41)$$

Finally, the attenuation coefficient is calculated as

$$\hat{\beta} = \gamma_1 \cdot \gamma_2 \cdot \boldsymbol{a}^{\dagger}(\hat{\theta},\hat{d})\hat{\boldsymbol{g}}_1, \quad (42)$$

where $\gamma_1 = \boldsymbol{u}^{\dagger}(\hat{d})\hat{\boldsymbol{g}}_2$ and $\gamma_2 = (\boldsymbol{f}(\hat{v}^{\rho}) \odot \boldsymbol{S}\boldsymbol{a}(\hat{\theta},\hat{d}))^{\dagger}\hat{\boldsymbol{g}}_3$ are the scaling ambiguities associated with $\hat{\boldsymbol{g}}_2$ and $\hat{\boldsymbol{g}}_3$, respectively.

While this stage yields a computationally efficient coarse estimate, the imposed low-rank approximation restricts its capacity for high-precision localization and velocity estimation. These limitations motivate the refinement stage in Section IV-B, where the full parameter set is recovered with improved accuracy.

B. Parameter Refinement via Quasi-Newton Optimization

To further refine the estimation from the CP-based initialization, the quasi-Newton framework is incorporated to minimize the optimization function $J(\boldsymbol{\eta})$ in (34), which fully accounts for the nonlinear spatial-time coupling and improves estimation accuracy.

1) Configurations of BFGS Optimization: To facilitate iterative refinement of $\boldsymbol{\eta}$, the BFGS algorithm is employed for quasi-Newton optimization, owing to its robustness to estimation errors in the initialization stage. The parameter vector is initialized as $\boldsymbol{\eta}_0 = [\hat{d}, \hat{\theta}, \hat{v}^{\rho}, \hat{v}^{\phi}, \Re(\hat{\beta}), \Im(\hat{\beta})]^{\mathrm{T}}$ with $\hat{v}^{\phi}$ set to 0, and the inverse Hessian approximation $\boldsymbol{H}_0$ is set to the identity matrix $\mathbf{I}$.

At the t-th iteration, the parameter vector is presented as $\boldsymbol{\eta}_t = [d_t, \theta_t, v_t^{\rho}, v_t^{\phi}, \Re(\beta_t), \Im(\beta_t)]^{\mathrm{T}}$, and the approximated inverse Hessian matrix is denoted by $\boldsymbol{H}_t$. The gradient of the optimization function $J(\boldsymbol{\eta}_t)$ is calculated as

$$\nabla_{\boldsymbol{\eta}} J(\boldsymbol{\eta}_t) = \left[\frac{\partial J}{\partial d}, \frac{\partial J}{\partial \theta}, \frac{\partial J}{\partial v^{\rho}}, \frac{\partial J}{\partial v^{\phi}}, \frac{\partial J}{\partial \Re(\beta)}, \frac{\partial J}{\partial \Im(\beta)}\right]^{\mathrm{T}}. \quad (43)$$

Note that, the non-zero partial derivative $\frac{\partial J}{\partial v^{\phi}}$, often overlooked in the existing literature [12], [46], captures the influence of v^{ϕ} on the phase structure of $\boldsymbol{\mu}_{k,m}$ and introduces observability of v^{ϕ} into the optimization process.

According to the definition of J in (34), each element in $\nabla_{\boldsymbol{\eta}} J(\boldsymbol{\eta}_t)$ is given by

$$\frac{\partial J}{\partial \eta} = \sum_{m,k} -2\Re\left\{\left\langle \mathbf{y}_{k,m} - \boldsymbol{\mu}_{k,m}, \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \eta}\right\rangle\right\}, \quad \forall \eta \in \boldsymbol{\eta}, \quad (44)$$

where the derivatives $\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \eta}$ are detailed in Appendix A.

2) Iterative Updates and Convergence Criteria: In each iteration, the parameter vector $\boldsymbol{\eta}_t$ and the approximated inverse Hessian $\boldsymbol{H}_t$ are respectively updated as

$$\boldsymbol{\eta}_{t+1} = \boldsymbol{\eta}_t - \alpha_t \boldsymbol{H}_t \nabla_{\boldsymbol{\eta}} J(\boldsymbol{\eta}_t) \quad (45)$$

and

$$\boldsymbol{H}_{t+1} = \boldsymbol{H}_t + \frac{(\boldsymbol{\omega}_t^{\mathrm{T}}\boldsymbol{\varsigma}_t + \boldsymbol{\varsigma}_t^{\mathrm{T}}\boldsymbol{H}_t\boldsymbol{\varsigma}_t)(\boldsymbol{\omega}_t\boldsymbol{\omega}_t^{\mathrm{T}})}{(\boldsymbol{\omega}_t^{\mathrm{T}}\boldsymbol{\varsigma}_t)^2} - \frac{\boldsymbol{H}_t\boldsymbol{\varsigma}_t\boldsymbol{\omega}_t^{\mathrm{T}} + \boldsymbol{\omega}_t\boldsymbol{\varsigma}_t^{\mathrm{T}}\boldsymbol{H}_t}{\boldsymbol{\omega}_t^{\mathrm{T}}\boldsymbol{\varsigma}_t}, \quad (46)$$

where α_t denotes the optimal step size satisfying the Wolfe conditions [49] as

$$\begin{aligned} &J(\boldsymbol{\eta}_t - \alpha_t \boldsymbol{H}_t \nabla J(\boldsymbol{\eta}_t)) \le J(\boldsymbol{\eta}_t) - c_1\alpha_t \nabla J(\boldsymbol{\eta}_t)^{\mathrm{T}}\boldsymbol{H}_t\nabla J(\boldsymbol{\eta}_t) \\ &-\nabla J(\boldsymbol{\eta}_t - \alpha_t\boldsymbol{H}_t\nabla J(\boldsymbol{\eta}_t))^{\mathrm{T}}\boldsymbol{H}_t\nabla J(\boldsymbol{\eta}_t) \ge -c_2\nabla J(\boldsymbol{\eta}_t)^{\mathrm{T}}\boldsymbol{H}_t\nabla J(\boldsymbol{\eta}_t) \end{aligned} \quad (47)$$

with $0 < c_1 < c_2 < 1$, and $\boldsymbol{\omega}_t = \boldsymbol{\eta}_{t+1} - \boldsymbol{\eta}_t$, $\boldsymbol{\varsigma}_t = \nabla J(\boldsymbol{\eta}_{t+1}) - \nabla J(\boldsymbol{\eta}_t)$. The iterative refinement terminates when either gradient magnitude falls below a predefined threshold ϵ, i.e., $\|\nabla J(\boldsymbol{\eta}_t)\| < \epsilon$, indicating convergence to a local minimum, or the maximum number of iterations $T_{\max}$ is reached, ensuring computational tractability. The optimized parameter vector derived from the convergence criteria is denoted by $\boldsymbol{\eta}_{\mathrm{opt}}$.

Through this iterative process, the transverse velocity v^{ϕ} is jointly estimated alongside refinement to $\{d, \theta, v^{\rho}, \beta\}$, effectively compensating for the model simplifications in Section IV-A.

C. Computational Complexity Analysis

The computational complexity of the proposed algorithm is associated with coarse estimation and refinement in two stages, respectively. In the first stage, the computational complexity is dominated by the ALS algorithm, where each ALS iteration requires a complexity of $\mathcal{O}(NMK)$ to update the factor vectors. Thus, the total complexity for this stage is $\mathcal{O}(NMKN_{\mathrm{ALS}})$, where N_{ALS} denotes the number of iterations for the ALS algorithm to converge. For the second stage, the complexity is dominated by the gradient computation of the optimization function $J(\boldsymbol{\eta})$ in (34). This requires evaluating the NMK elements in each iteration, leading to a complexity of $\mathcal{O}(NMK)$. The complexity for this stage is $\mathcal{O}(NMKN_{\mathrm{BFGS}})$, where N_{BFGS} denotes the number of iterations for the BFGS optimization. Thus, the total computational complexity of the proposed algorithm is given as $\mathcal{O}(NMK \times (N_{\mathrm{ALS}} + N_{\mathrm{BFGS}}))$.

V. Simulation Results

In this section, numerical results are provided to characterize the performance bounds and validate the effectiveness of the proposed algorithm in the near-field high-mobility scenario. The BS is equipped with $N = 128$ antenna elements, resulting in an array aperture of $D = (N-1)\lambda/2 = 0.68$ m. This closely matches the aperture dimensions of existing commercial BSs [50]. For this configuration, the corresponding Fresnel near-field region spans from $0.62\sqrt{D^3/\lambda} \approx 3.4$ m to $2D^2/\lambda \approx 86.4$ m. The carrier frequency, subcarrier spacing, and the symbol period of the OFDM waveforms are set to $f_c = 28$ GHz, $\Delta = 120$ kHz, and $T_{\mathrm{s}} = 8.92$ μs, respectively, in accordance with the 3GPP Technical Specification [51]. Quadrature phase-shift keying (QPSK) modulation is employed for OFDM symbol generation. Unless otherwise

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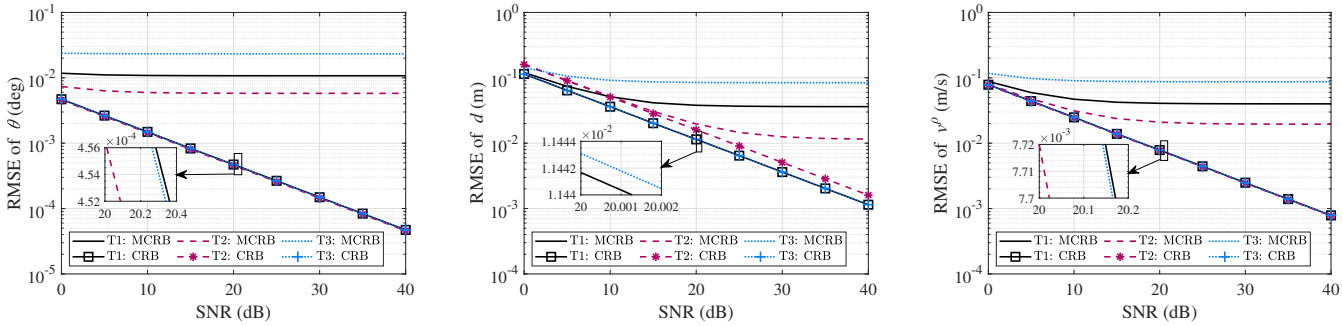

(a) (b) (c)

Fig. 3: Performance bounds under model misspecification for (a) DOA θ; (b) Range d; (c) Radial velocity v^{ρ} corresponding to three targets: T1, T2, and T3 with parameters listed in TABLE II.

TABLE II: Target Parameter Configurations

Target	$\bar{d}$ [m]	$\bar{v}^{\phi}$ [m/s]	$\bar{\theta}$ [°]	$\bar{v}^{\rho}$ [m/s]
T1	13.4235	24.2243	32.2323	10.5485
T2	**24.7335**	24.2243	32.2323	10.5485
T3	13.4235	**52.2355**	32.2323	10.5485

specified, the number of subcarriers and OFDM symbols are set to $K = 16$ and $M = 16$, respectively.

The simulated target is located within the Fresnel near-field region of the BS. The range, direction, radial velocity and transverse velocity of the target are set to $\bar{d} = 13.4235$ m, $\bar{\theta} = 32.2323°$, $\bar{v}^{\rho} = 10.5485$ m/s, and $\bar{v}^{\phi} = 24.2243$ m/s, respectively. The attenuation coefficient is set to $\bar{\beta} = -0.8283 + \jmath 0.1273$. The accuracy of parameter estimation is assessed using the root mean square error (RMSE) metric. For each simulated data point, 500 Monte Carlo trials are run.

A. Performance Bounds

To validate the theoretical analysis of $\mathrm{CRB}(\bar{\boldsymbol{\eta}})$ and $\mathrm{MCRB}(\bar{\boldsymbol{\eta}})$, we analyze the performance bounds under model misspecification with respect to SNR, range $\bar{d}$, and transverse velocity $\bar{v}^{\phi}$. To ensure consistent units for theoretical bounds and estimation errors, the CRBs and MCRBs are normalized by applying a square-root scaling to the matrices in (24) and (32), respectively.

1) Performance Bounds v.s. SNR: In Fig. 3, we present the performance bounds for DOA, range and radial velocity estimation corresponding to three targets, T1, T2, and T3, whose specific parameters are summarized in Table II. It is evident that the MCRBs consistently exceed the corresponding CRBs, which aligns with the analysis in Section III-B. Moreover, as the SNR increases, the disparity between the MCRB and the CRB becomes larger. This is because the systematic bias resulting from model misspecification becomes the dominant component of estimation error at higher SNR levels.

Furthermore, comparing the performance bounds across different targets reveals that a smaller range and a larger transverse velocity exacerbate the discrepancy between the MCRB and the CRB. In other words, near-field targets with high transverse velocities are more challenging to characterize accurately using misspecified models.

2) Performance Bounds v.s. Range and Transverse Velocity: To quantify the impact of range and transverse velocity on model misspecification, we use DOA estimation as an example, evaluating the performance bounds under varying $\bar{d}$ and $\bar{v}^{\phi}$ in Fig. 5. It is clear in Fig. 5(a) that the MCRB decreases consistently with increasing target range. Meanwhile, the MCRB increases with larger transverse velocity, as shown in Fig. 5(b). These results are consistent with the observations in Fig. 3 with various target parameters.

Moreover, the impact of the number of subcarriers K and symbol durations M on the performance bounds is also demonstrated in Fig. 5. As discussed in *Remark 2*, increases in K and M do not necessarily yield lower MCRBs, since the estimation bias caused by model misspecification also increases with more resources in either the frequency or time domain.

B. Estimation Accuracy

In the following, we compare the estimation accuracy of the proposed algorithm with that of the TeNFiLoc method [12], fractional Fourier transform (FrFT)-based method [32], joint angle-of-departure and Doppler shift estimation (JADE) method [40], and the least squares (LS)-based method [46].

In particular, the TeNFiLoc, JADE, and LS-based methods employ tensor decomposition processing, whereas the FrFT-based method utilizes a two-stage approach consisting of a coarse parameter estimation via FrFT followed by Newton-based refinement. For the TeNFiLoc, JADE, LS-based methods and the initialization stage of the proposed algorithm, the CP decomposition is implemented using Tensorlab 3.0 [52]. While velocity estimation is not included in the LS-based method, it is extended using the same methodology as the proposed algorithm. The FrFT-based method employs a 512-point zero-padding in the time domain and a 1024-point zero-padding in the frequency domain. For the refinement stage of FrFT-based method and the proposed algorithm, the optimization is performed via the `fminunc` and `fmincon` functions in MATLAB, respectively. The CRBs for the estimation of various parameters are presented for reference.

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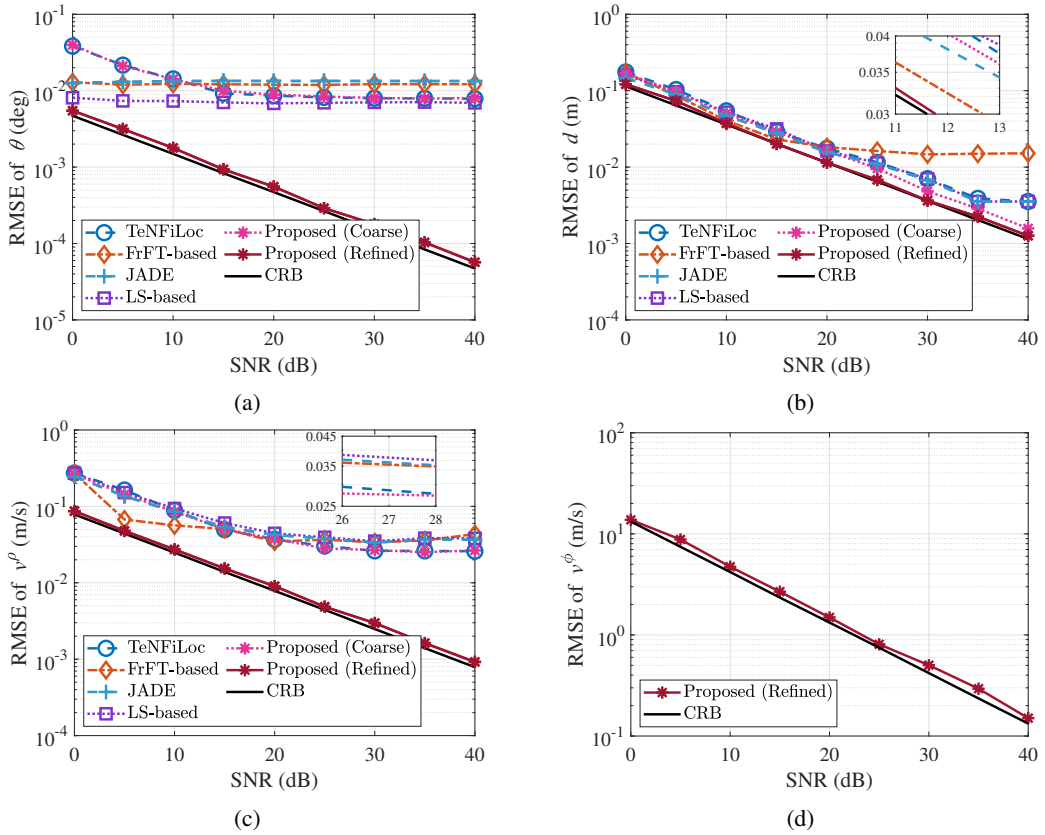

Fig. 4: Comparison of estimation accuracy *versus* SNR for (a) DOA θ; (b) Range d; (c) Radial velocity v^{ρ}; (d) Transverse velocity v^{ϕ}.

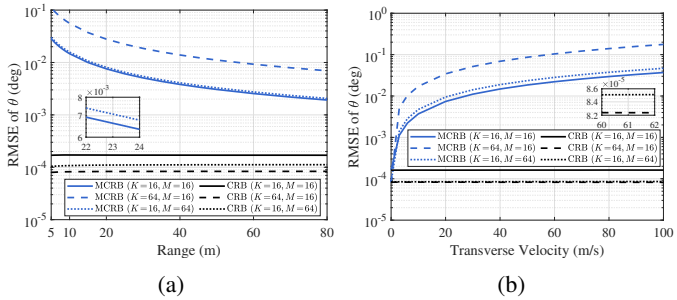

Fig. 5: Performance bounds under model misspecification for DOA θ with varying (a) Range $\bar{d}$; (b) Transverse velocity $\bar{v}^{\phi}$.

To demonstrate the effectiveness of the proposed two-stage algorithm, we present the simulation results from both stages. The coarse estimate from the first stage is labeled as "Proposed (Coarse)", and the refined estimate from the second stage is labeled as "Proposed (Refined)". This allows for a clear visualization of how the initial coarse estimation provides a solid foundation for the subsequent refinement stage, which in turn enhances the estimation accuracy.

1) Estimation Accuracy v.s. SNR: It is observed from Fig. 4 that the coarse estimation in the first stage achieves an accuracy comparable to those of the compared methods, while the refined estimation clearly outperforms all the compared methods in estimating DOA θ, range d, and radial velocity v^{ρ} across the entire SNR range. This superior performance stems from its reliance on the exact near-field high-mobility model, which accurately captures the Doppler shift induced by transverse velocity. In contrast, the compared methods rely on simplified models that assume Doppler shifts are solely associated with radial velocity.

Moreover, Fig. 4(d) presents the estimation results for the transverse velocity v^{ϕ}. It is important to note that the compared methods do not provide estimates for v^{ϕ} because their signal models neglect the Doppler shift induced by transverse motion. Although the signal model could be modified to incorporate transverse velocity, the existing methods are not directly applicable to the revised model for two primary reasons. First, the TeNFiLoc, JADE, and LS-based methods rely on low-rank tensor decomposition, while introducing v^{ϕ} into the signal model disrupts the inherent low-rankness and prevents an effective low-rank decomposition. Second, for the FrFT-based method, the inclusion of v^{ϕ} leads to nonlinear frequency modulation in the phase term, which undermines the performance of the FrFT that is intended for linear chirp signals.

2) Estimation Accuracy v.s. Range: In Fig. 6, we evaluate the parameter estimation accuracy by changing the range $\bar{d}$ between the target and the array center. It is clear that the proposed algorithm demonstrates the highest accuracy within the evaluated near-field region. Specifically, while existing methods exhibit gradually decreasing errors in DOA and radial velocity estimation with increasing range, their estimation errors at a range of 100 m still exceed those of the proposed algorithm. This persistent performance gap

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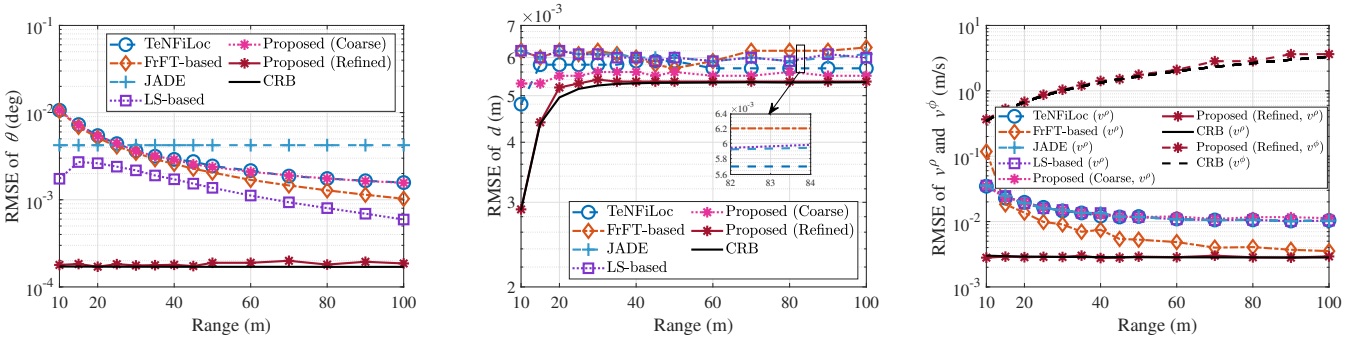
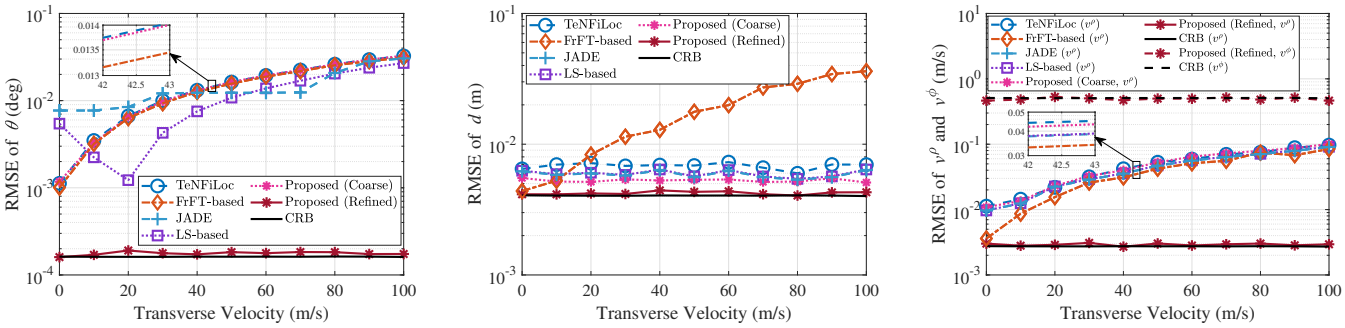


(a) (b) (c)

Fig. 6: Comparison of estimation accuracy versus range $\bar{d}$ for (a) DOA θ; (b) Range d; (c) Radial velocity v^{ρ} and transverse velocity v^{ϕ}, where SNR $= 30$ dB.



(a) (b) (c)

Fig. 7: Comparison of estimation accuracy versus transverse velocity $\bar{v}^{\phi}$ for (a) DOA θ; (b) Range d; (c) Radial velocity v^{ρ} and transverse velocity v^{ϕ}, where SNR $= 30$ dB.

highlights the superiority of the proposed algorithm for near-field high-mobility targets. Moreover, the proposed algorithm is inherently applicable to far-field conditions. This is because the plane wave model, commonly used for the far-field, is a well-established approximation of the more general spherical wave model upon which our method is based.

Meanwhile, it is demonstrated in Fig. 6(c) that the estimation accuracy of the transverse velocity decreases with increasing distance, following a similar trend observed in the corresponding CRB. This is because as the distance increases, the differences in the projection components ν_n^{ϕ} in (3) of the transverse velocity across the array elements decrease. As a result, it becomes increasingly challenging to capture the influence of the transverse velocity on the received signal in both the temporal and spatial domains, thereby complicating its accurate estimation.

3) Estimation Accuracy v.s. Transverse Velocity: Finally, the parameter estimation accuracy with varying transverse velocity $\bar{v}^{\phi}$ is compared in Fig. 7. It is clear that the proposed algorithm enjoys an enhanced accuracy in the estimation of DOA θ, range d and radial velocity v^{ρ} for different transverse velocities $\bar{v}^{\phi}$. This robustness benefits from the two-stage methodology. The first stage provides reliable coarse estimates, and the second stage effectively compensates for the Doppler effects of transverse motion using the exact model, resulting in stable and accurate estimates. Moreover, whereas the performance of the proposed algorithm remains nearly unchanged when $\bar{v}^{\phi}$ varies, the compared methods show increasingly larger estimation errors, particularly for DOA θ and radial velocity v^{ρ}. This is attributed to the growing contribution of the term $e^{\jmath\frac{2\pi}{\lambda}\nu_n^{\phi} m T_{\mathrm{s}}}$ in $e^{\jmath\frac{2\pi}{\lambda}\nu_n m T_{\mathrm{s}}}$ for the Doppler shift vector in (8) as the transverse velocity v^{ϕ} increases, leading to a greater deviation when v^{ϕ} is omitted in signal modeling.

VI. Conclusions

In this paper, we analyze the critical impact of neglecting transverse velocity on parameter estimation accuracy for near-field high-mobility targets and propose an efficient algorithm to address this challenge. By integrating low-rank tensor approximation with quasi-Newton refinement, the proposed algorithm effectively estimates and refines both location and velocity parameters, improving localization accuracy for high-mobility targets in near-field environments. Simulation results demonstrate that the proposed algorithm consistently outperforms conventional methods. This work bridges a gap in near-field sensing technology for ISAC implementation by capturing full motion dynamics of non-cooperative targets.

Appendix A
Derivation of Entries in the CRB

To derive the CRB, we compute the first-order partial derivatives of $\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})$ with respect to each parameter in $\boldsymbol{\eta}$.

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A. First-Order Derivatives with Respect to β

From the noise-less signal model $\boldsymbol{\mu}_{k,m}$ in (12), the partial derivatives with respect to the real and imagination parts of the attenuation factor β are derived as

$$\begin{aligned}\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}}} &= e^{-\jmath 2\pi k\Delta\frac{2d}{c}}\boldsymbol{A}(\theta,d)\odot\boldsymbol{D}_m(\theta,d,v^{\rho},v^{\phi})\mathbf{s}_m=\frac{\boldsymbol{\mu}_{k,m}}{\beta},\\ \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{I}}} &= \jmath e^{-\jmath 2\pi k\Delta\frac{2d}{c}}\boldsymbol{A}(\theta,d)\odot\boldsymbol{D}_m(\theta,d,v^{\rho},v^{\phi})\mathbf{s}_m=\jmath\frac{\boldsymbol{\mu}_{k,m}}{\beta},\end{aligned} \tag{48}$$

respectively.

B. First-Order Derivatives with Respect to d

Let $\tilde{\beta}=\beta e^{-\jmath 2\pi k\Delta\frac{2d}{c}}$, the partial derivative with respect to d becomes

$$\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial d} = \tilde{\beta}\left[-\frac{\jmath 4\pi k\Delta}{c}\boldsymbol{A}\odot\boldsymbol{D}_m+\dot{\boldsymbol{A}}^{d}\odot\boldsymbol{D}_m+\boldsymbol{A}\odot\dot{\boldsymbol{D}}_m^{d}\right]\mathbf{s}_m, \tag{49}$$

where

$$\begin{aligned}\dot{\boldsymbol{A}}^{d} \triangleq \frac{\partial \boldsymbol{A}}{\partial d} &= \left(\frac{\partial \boldsymbol{a}}{\partial d}\right)\boldsymbol{a}^{\mathrm{T}}+\boldsymbol{a}\left(\frac{\partial \boldsymbol{a}^{\mathrm{T}}}{\partial d}\right)\\ &= -\jmath\frac{2\pi}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}}{\partial d}\right)\boldsymbol{A}+\boldsymbol{A}\mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}}{\partial d}\right)\right)\end{aligned} \tag{50}$$

and

$$\begin{aligned}\dot{\boldsymbol{D}}_m^{d} \triangleq \frac{\partial \boldsymbol{D}_m}{\partial d} &= \left(\frac{\partial \boldsymbol{d}_m}{\partial d}\right)\boldsymbol{d}_m^{\mathrm{T}}+\boldsymbol{d}_m\left(\frac{\partial \boldsymbol{d}_m^{\mathrm{T}}}{\partial d}\right)\\ &= \jmath\frac{2\pi m T_{\mathrm{s}}}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial d}\right)\boldsymbol{D}_m+\boldsymbol{D}_m\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial d}\right)\right)\end{aligned} \tag{51}$$

denote respectively the first derivatives of $\boldsymbol{A}$ and $\boldsymbol{D}_m$ with respect to d, and $\boldsymbol{\delta}=[\delta_1,\delta_2,\cdots,\delta_N]^{\mathrm{T}}$. For notation simplicity, we use $\dot{(\cdot)}^{\eta}$ to denote the first-order derivative with respect to η throughout this paper. Substituting (6) into $\frac{\partial\boldsymbol{\delta}}{\partial d}$, the n-th element of $\frac{\partial\boldsymbol{\delta}}{\partial d}$ becomes

$$\frac{\partial \delta_n}{\partial d} = \frac{2d-\dot{n}\lambda\cos\theta}{2r_n}-1. \tag{52}$$

Similarly, substituting (2) and (3) into $\frac{\partial\boldsymbol{\nu}}{\partial d}$ yields

$$\frac{\partial \nu_n}{\partial d} = v^{\rho}\frac{\dot{n}^2\lambda^2\sin^2\theta}{4r_n^3}-v^{\phi}\frac{\dot{n}\lambda\sin\theta\big(2d-\dot{n}\lambda\cos\theta\big)}{4r_n^3}. \tag{53}$$

C. First-Order Derivatives with Respect to θ

The partial derivative of $\boldsymbol{\mu}_{k,m}$ with respect to θ is expressed as

$$\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \theta} = \tilde{\beta}\left[\dot{\boldsymbol{A}}^{\theta}\odot\boldsymbol{D}_m+\boldsymbol{A}\odot\dot{\boldsymbol{D}}_m^{\theta}\right]\mathbf{s}_m, \tag{54}$$

where

$$\dot{\boldsymbol{A}}^{\theta}\triangleq\frac{\partial \boldsymbol{A}}{\partial \theta}=-\jmath\frac{2\pi}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}}{\partial \theta}\right)\boldsymbol{A}+\boldsymbol{A}\,\mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}}{\partial \theta}\right)\right), \tag{55}$$

and

$$\dot{\boldsymbol{D}}_m^{\theta}\triangleq\frac{\partial \boldsymbol{D}_m}{\partial \theta}=\jmath\frac{2\pi m T_{\mathrm{s}}}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial \theta}\right)\boldsymbol{D}_m+\boldsymbol{D}_m\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial \theta}\right)\right). \tag{56}$$

By substituting (6) into $\frac{\partial\boldsymbol{\delta}}{\partial\theta}$ and substituting (2) and (3) into $\frac{\partial\boldsymbol{\nu}}{\partial\theta}$, we have

$$\frac{\partial \delta_n}{\partial \theta} = \frac{\dot{n}d\lambda\sin\theta}{2r_n}, \tag{57}$$

$$\frac{\partial \nu_n}{\partial \theta} = \frac{\dot{n}^3\lambda^3\sin\theta}{8r_n^3}v^{\rho}+\left[\frac{\dot{n}\lambda\cos\theta}{2r_n}-\frac{d\,\dot{n}^2\lambda^2\sin^2\theta}{4r_n^3}\right]v^{\phi}, \tag{58}$$

respectively.

D. First-Order Derivatives with Respect to v^{ρ} and v^{ϕ}

For radial and transverse velocity parameters, the partial derivatives are

$$\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^{\rho}} = \tilde{\beta}\left[\boldsymbol{A}\odot\dot{\boldsymbol{D}}_m^{v^{\rho}}\right]\mathbf{s}_m \tag{59}$$

and

$$\frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^{\phi}} = \tilde{\beta}\left[\boldsymbol{A}\odot\dot{\boldsymbol{D}}_m^{v^{\phi}}\right]\mathbf{s}_m, \tag{60}$$

where

$$\begin{aligned}\dot{\boldsymbol{D}}_m^{v^{\rho}} &\triangleq \frac{\partial \boldsymbol{D}_m}{\partial v^{\rho}} = \jmath\frac{2\pi m T_{\mathrm{s}}}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial v^{\rho}}\right)\boldsymbol{D}_m+\boldsymbol{D}_m\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial v^{\rho}}\right)\right),\\ \dot{\boldsymbol{D}}_m^{v^{\phi}} &\triangleq \frac{\partial \boldsymbol{D}_m}{\partial v^{\phi}} = \jmath\frac{2\pi m T_{\mathrm{s}}}{\lambda}\left(\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial v^{\phi}}\right)\boldsymbol{D}_m+\boldsymbol{D}_m\mathrm{diag}\left(\frac{\partial \boldsymbol{\nu}}{\partial v^{\phi}}\right)\right).\end{aligned} \tag{61}$$

By substituting (3) into $\frac{\partial\boldsymbol{\nu}}{\partial v^{\rho}}$ and $\frac{\partial\boldsymbol{\nu}}{\partial v^{\phi}}$, we have

$$\frac{\partial \nu_n}{\partial v^{\rho}} = \frac{2d-\dot{n}\lambda\cos\theta}{2r_n},\quad \frac{\partial \nu_n}{\partial v^{\phi}} = \frac{\dot{n}\lambda\sin\theta}{2r_n}, \tag{62}$$

respectively.

Appendix B
Derivation of Entries in the MCRB

By substituting (11) into (30) and (31), the (i,j)-th element in $\boldsymbol{C}_{\boldsymbol{\eta}^*}$ and $\boldsymbol{B}_{\boldsymbol{\eta}^*}$ are respectively written as

$$\begin{aligned}[\boldsymbol{C}(\boldsymbol{\eta}^*)]_{(i,j)} &= -\int\frac{\partial^2\ln\tilde{p}(\boldsymbol{y}\mid\boldsymbol{\eta})}{\partial\eta_i\partial\eta_j}p(\boldsymbol{y}\mid\bar{\boldsymbol{\eta}})\,\mathrm{d}\boldsymbol{y}\Bigg|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}\\ &= \frac{2}{\sigma^2}\sum_{k,m}\Re\left\{\boldsymbol{\varepsilon}_{k,m}^{\mathrm{H}}(\boldsymbol{\eta})\frac{\partial^2\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})}{\partial\eta_i\partial\eta_j}\right.\\ &\qquad\left.+\left(\frac{\partial\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})}{\partial\eta_i}\right)^{\mathrm{H}}\frac{\partial\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})}{\partial\eta_j}\right\}\Bigg|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}\end{aligned} \tag{63}$$

and

$$\begin{aligned}[\boldsymbol{B}(\boldsymbol{\eta}^*)]_{(i,j)} &= \int\frac{\partial\ln\tilde{p}(\boldsymbol{y}\,|\,\boldsymbol{\eta})}{\partial\eta_i}\frac{\partial\ln\tilde{p}(\boldsymbol{y}\,|\,\boldsymbol{\eta})}{\partial\eta_j}p(\boldsymbol{y}\,|\,\bar{\boldsymbol{\eta}})\,d\boldsymbol{y}\Bigg|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}\\ &= \frac{2}{\sigma^2}\sum_{k,m}\left[\frac{2}{\sigma^2}\Re\left\{\boldsymbol{\varepsilon}_{k,m}^{\mathrm{H}}\frac{\partial\boldsymbol{\mu}_{k,m}}{\partial\eta_i}\right\}\Re\left\{\boldsymbol{\varepsilon}_{k,m}^{\mathrm{H}}\frac{\partial\boldsymbol{\mu}_{k,m}}{\partial\eta_j}\right\}\right.\\ &\qquad\left.+\Re\left\{\left(\frac{\partial\boldsymbol{\mu}_{k,m}}{\partial\eta_i}\right)^{\mathrm{H}}\frac{\partial\boldsymbol{\mu}_{k,m}}{\partial\eta_j}\right\}\right]\Bigg|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*},\end{aligned} \tag{64}$$

where η_i and η_j denote the i-th and j-th parameter in the vector $\boldsymbol{\eta}=[d,\theta,v^{\rho},v^{\phi},\beta_{\mathrm{R}},\beta_{\mathrm{I}}]^{\mathrm{T}}$, and $\boldsymbol{\varepsilon}_{k,m}(\boldsymbol{\eta})=\boldsymbol{\mu}_{k,m}(\boldsymbol{\eta})-\boldsymbol{\mu}_{k,m}(\bar{\boldsymbol{\eta}})$. The computation of $\boldsymbol{C}(\boldsymbol{\eta}^*)$ and $\boldsymbol{B}(\boldsymbol{\eta}^*)$ necessitates both first- and second-order derivatives of $\boldsymbol{\mu}_{k,m}$. While the

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first-order derivatives are given in Appendix A, this appendix focuses on the second-order derivatives.

Based on Schwarz's theorem on the symmetry of mixed partial derivatives, the equality $\frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \eta_i \partial \eta_j} = \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \eta_j \partial \eta_i}$ holds under continuity condition of the second-order derivatives. This reduces the number of unique cross-derivative computations by half. In the subsequent derivation, we present the non-zero second-order derivatives of $\boldsymbol{\mu}_{k,m}$ with respect to parameter pairs $(\eta_i, \eta_j) \in \boldsymbol{\eta} \times \boldsymbol{\eta}$.

A. Second-Order Derivatives with Respect to $\eta_i \in \{\beta_{\mathrm{R}}, \beta_{\mathrm{I}}\}$ and $\eta_j \in \{d, \theta, v^\rho, v^\phi, \beta_{\mathrm{R}}, \beta_{\mathrm{I}}\}$

The second-order derivatives with respect to the amplitude parameters β_{R} and β_{I} are

$$\frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial (\beta_{\mathrm{R}})^2} = 0, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial (\beta_{\mathrm{I}})^2} = 0, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}} \partial \beta_{\mathrm{I}}} = 0. \tag{65}$$

For cross-derivatives between the attenuation coefficient β_{R} and other location and velocity parameters, we have

$$\begin{aligned} \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}} \partial d} &= \frac{1}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial d}, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}} \partial \theta} = \frac{1}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \theta}, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}} \partial v^\rho} &= \frac{1}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^\rho}, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{R}} \partial v^\phi} = \frac{1}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^\phi}. \end{aligned} \tag{66}$$

The corresponding derivatives with β_{I} follow similar patterns through complex conjugation as

$$\begin{aligned} \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{I}} \partial d} &= \frac{\jmath}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial d}, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{I}} \partial \theta} = \frac{\jmath}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial \theta}, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{I}} \partial v^\rho} &= \frac{\jmath}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^\rho}, \quad \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \beta_{\mathrm{I}} \partial v^\phi} = \frac{\jmath}{\beta} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial v^\phi}. \end{aligned} \tag{67}$$

B. Second-Order Derivatives with Respect to $\eta_i, \eta_j \in \{d, \theta\}$

The second-order partial derivatives involving the location parameters range d and DOA θ are represented as

$$\begin{aligned} \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial d^2} &= -\frac{\jmath 4\pi k \Delta}{c} \frac{\partial \boldsymbol{\mu}_{k,m}}{\partial d} + \tilde{\beta} \Big[-\frac{\jmath 4\pi k \Delta}{c} \Big(\dot{\boldsymbol{A}}^d \odot \boldsymbol{D}_m + \boldsymbol{A} \odot \dot{\boldsymbol{D}}_m^d \Big) \\ &\quad + \ddot{\boldsymbol{A}}^{d,d} \odot \boldsymbol{D}_m + 2 \dot{\boldsymbol{A}}^d \odot \dot{\boldsymbol{D}}_m^d + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{d,d} \Big] \mathbf{s}_m, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial d \partial \theta} &= \tilde{\beta} \Big[-\frac{\jmath 4\pi k \Delta}{c} (\dot{\boldsymbol{A}}^\theta \odot \boldsymbol{D}_m + \boldsymbol{A} \odot \dot{\boldsymbol{D}}_m^\theta) + \ddot{\boldsymbol{A}}^{d,\theta} \odot \boldsymbol{D}_m \\ &\quad + \dot{\boldsymbol{A}}^d \odot \dot{\boldsymbol{D}}_m^\theta + \dot{\boldsymbol{A}}^\theta \odot \dot{\boldsymbol{D}}_m^d + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{d,\theta} \Big] \mathbf{s}_m, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial d \theta^2} &= \tilde{\beta} \Big[\ddot{\boldsymbol{A}}^{\theta,\theta} \odot \boldsymbol{D}_m + 2 \dot{\boldsymbol{A}}^\theta \odot \dot{\boldsymbol{D}}_m^\theta + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{\theta,\theta} \Big] \mathbf{s}_m, \end{aligned} \tag{68}$$

where

$$\begin{aligned} \ddot{\boldsymbol{A}}^{\eta_i,\eta_j} &\triangleq \frac{\partial^2 \boldsymbol{A}}{\partial \eta_i \partial \eta_j} = -\jmath \frac{2\pi}{\lambda} \Bigg[\mathrm{diag}\Big(\frac{\partial^2 \boldsymbol{\delta}}{\partial \eta_i \partial \eta_j} \Big) \boldsymbol{A} + \mathrm{diag}\Big(\frac{\partial \boldsymbol{\delta}}{\partial \eta_i} \Big) \dot{\boldsymbol{A}}^{\eta_j} \\ &\quad + \dot{\boldsymbol{A}}^{\eta_j} \mathrm{diag}\left(\frac{\partial \boldsymbol{\delta}}{\partial \eta_i} \right) + \boldsymbol{A} \mathrm{diag}\left(\frac{\partial^2 \boldsymbol{\delta}}{\partial \eta_i \partial \eta_j} \right) \Bigg] \end{aligned} \tag{69}$$

and

$$\begin{aligned} \ddot{\boldsymbol{D}}_m^{\eta_i,\eta_j} &\triangleq \frac{\partial^2 \boldsymbol{D}_m}{\partial \eta_i \partial \eta_j} = \jmath \frac{2\pi m T_{\mathrm{s}}}{\lambda} \Bigg[\mathrm{diag}\Big(\frac{\partial^2 \boldsymbol{\nu}}{\partial \eta_i \partial \eta_j} \Big) \boldsymbol{D}_m + \mathrm{diag}\Big(\frac{\partial \boldsymbol{\nu}}{\partial \eta_i} \Big) \dot{\boldsymbol{D}}_m^{\eta_j} \\ &\quad + \dot{\boldsymbol{D}}_m^{\eta_j} \mathrm{diag}\Big(\frac{\partial \boldsymbol{\nu}}{\partial \eta_i} \Big) + \boldsymbol{D}_m \mathrm{diag}\Big(\frac{\partial^2 \boldsymbol{\nu}}{\partial \eta_i \partial \eta_j} \Big) \Bigg] \end{aligned} \tag{70}$$

denote the second-order derivatives of $\boldsymbol{A}$ and $\boldsymbol{D}_m$ with respect to η_i and η_j, respectively. In (69) and (70), the n-th element in the second-order derivatives of $\boldsymbol{\delta}$ and $\boldsymbol{\nu}$ with respect to $\eta_i, \eta_j \in \{d, \theta\}$ can be calculated by

$$\begin{aligned} \frac{\partial^2 \delta_n}{\partial d^2} &= \frac{\dot{n}^2 \lambda^2 \sin^2 \theta}{4 r_n^3}, \\ \frac{\partial^2 \delta_n}{\partial d \partial \theta} &= \frac{\dot{n}^2 \lambda^2 \sin \theta (\dot{n} \lambda - 2d \cos \theta)}{8 r_n^3}, \\ \frac{\partial^2 \delta_n}{\partial \theta^2} &= \frac{\dot{n} d \lambda \left[2 r_n^2 \cos \theta - \dot{n} d \lambda \sin^2 \theta \right]}{4 r_n^3}, \end{aligned} \tag{71}$$

and

$$\begin{aligned} \frac{\partial^2 \nu_n}{\partial d^2} &= \left(\frac{6(d - \frac{\dot{n}\lambda}{2} \cos \theta)^2}{2 r_n^5} - \frac{3}{2 r_n^3} \right) v^\rho \\ &\quad + \left(\frac{3 \dot{n} \lambda \sin \theta (2d - \dot{n} \lambda \cos \theta)}{4 r_n^5} \right) v^\phi, \\ \frac{\partial^2 \nu_n}{\partial d \partial \theta} &= \left(-\frac{\dot{n} \lambda \sin \theta}{2 r_n^3} + \frac{3 \dot{n} \lambda \sin \theta (2d - \dot{n} \lambda \cos \theta)^2}{2 r_n^5} \right) v^\rho \\ &\quad + \left(-\frac{\dot{n} \lambda \cos \theta}{2 r_n^3} + \frac{3 \dot{n}^2 \lambda^2 \sin^2 \theta (2d - \dot{n} \lambda \cos \theta)}{8 r_n^5} \right) v^\phi, \\ \frac{\partial^2 \nu_n}{\partial \theta^2} &= \frac{\dot{n} \lambda}{2 r_n^3} \Bigg[v^\rho \Bigg(r_n^2 \cos \theta - \frac{\dot{n} \lambda \sin^2 \theta}{2} \left(d - \frac{\dot{n} \lambda}{2} \cos \theta \right) \Bigg) \\ &\quad - v^\phi \Bigg(r_n^2 \sin \theta + \frac{3 \dot{n} \lambda \sin^2 \theta}{2} \left(d - \frac{\dot{n} \lambda}{2} \cos \theta \right) \Bigg) \Bigg] \\ &\quad + \frac{3 \lambda^2 \sin^2 \theta \left(r_n^2 - \frac{\dot{n}^2 \lambda^2}{4} \right)}{4 r_n^5} \left(d - \frac{\dot{n} \lambda}{2} \cos \theta \right) (v^\rho + v^\phi), \end{aligned} \tag{72}$$

respectively.

C. Second-Order Derivatives with Respect to $\eta_i \in \{d, \theta\}$ and $\eta_j \in \{v^\rho, v^\phi\}$

For the radial and transverse velocities, the second-order derivatives can be calculated by

$$\begin{aligned} \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial d \, \partial v^\rho} &= \tilde{\beta} \Big[-\frac{\jmath 4\pi k \Delta}{c} \boldsymbol{A} \odot \boldsymbol{D}_m^{v^\rho} + \dot{\boldsymbol{A}}^d \odot \dot{\boldsymbol{D}}_m^{v^\rho} + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{d,v^\rho} \Big] \mathbf{s}_m, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial d \, \partial v^\phi} &= \tilde{\beta} \Big[-\frac{\jmath 4\pi k \Delta}{c} \boldsymbol{A} \odot \dot{\boldsymbol{D}}_m^{v^\phi} + \dot{\boldsymbol{A}}^d \odot \dot{\boldsymbol{D}}_m^{v^\phi} + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{d,v^\phi} \Big] \mathbf{s}_m, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \theta \, \partial v^\rho} &= \tilde{\beta} \Big[\dot{\boldsymbol{A}}^\theta \odot \dot{\boldsymbol{D}}_m^{v^\rho} + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{\theta,v^\rho} \Big] \mathbf{s}_m, \\ \frac{\partial^2 \boldsymbol{\mu}_{k,m}}{\partial \theta \partial v^\phi} &= \tilde{\beta} \Big[\dot{\boldsymbol{A}}^\theta \odot \dot{\boldsymbol{D}}_m^{v^\phi} + \boldsymbol{A} \odot \ddot{\boldsymbol{D}}_m^{\theta,v^\phi} \Big] \mathbf{s}_m. \end{aligned} \tag{73}$$

Herein, to calculate $\ddot{\boldsymbol{A}}^{\eta_i,\eta_j}$ and $\ddot{\boldsymbol{D}}_m^{\eta_i,\eta_j}$ defined in (69) and (70), the second-order derivatives of ν_n with respect to $\eta_i \in \{d, \theta\}$

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and $\eta_j \in \{v^\rho, v^\phi\}$ can be respectively obtained by

$$\begin{aligned}
\frac{\partial^2 \nu_n}{\partial d \partial v^\rho} &= \frac{\dot{n}^2 \lambda^2 \sin^2\theta}{4 r_n^3}, \\
\frac{\partial^2 \nu_n}{\partial \theta \partial v^\rho} &= \frac{\dot{n}\lambda \sin\theta}{2 r_n} - \frac{\dot{n}^3 \lambda^3 \sin\theta}{8 r_n^3}, \\
\frac{\partial^2 \nu_n}{\partial d \partial v^\phi} &= -\frac{\dot{n}\lambda \sin\theta (2d - \dot{n}\lambda\cos\theta)}{4 r_n^3}, \\
\frac{\partial^2 \nu_n}{\partial \theta \partial v^\phi} &= \frac{\dot{n}\lambda \cos\theta}{2 r_n} - \frac{d\dot{n}^2 \lambda^2 \sin^2\theta}{4 r_n^3}.
\end{aligned} \tag{74}$$

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